1. Ders

$$z = x + iy = r\cdot[\cos\theta + i\sin\theta]$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \qquad e^{i\theta} = \sum_{n=0}^{\infty} \frac{(i\theta)^n}{n!} = \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!}\right) + i\left(\frac{\theta}{1!} - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} \ldots\right)$$

$$= \left(\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!}\theta^{2n}\right) + i\left(\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!}\theta^{(2n+1)}\right)$$

$$\boxed{e^{i\theta} = \cos\theta + i\sin\theta}$$

$$\theta = \pi \qquad e^{i\pi} = -1$$

$z = x + iy$ (cartesian)

$z = r[\cos\theta + i\sin\theta]$ (polar)

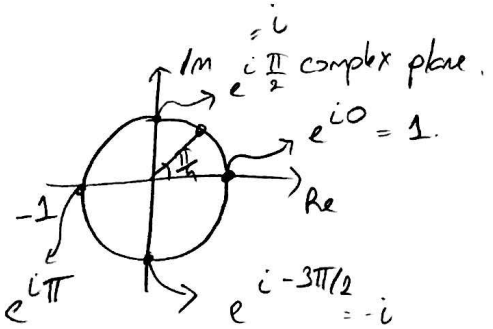
$z = re^{i\theta}$ (compacted with magnitude r, and angle θ) $r \geq 0$, θ - angle

Find: $-5e^{i\pi/2}$ in magnitude representation.

$r = 5 \qquad \theta = 3\pi/2$

$-5e^{-i\pi/2} = e^{i\pi}\cdot 5\, e^{i\pi/2} = 5\, e^{i\frac{3\pi}{2}}$, in general $-5e^{-i\pi/2} = e^{i(2n+1)\pi}\cdot 5\, e^{i\pi/2}$

$= 5\, e^{i\pi(2n+\frac{3}{2})}$



$$e^{-i\frac{\pi}{4}} = \frac{1}{\sqrt{2}}(1+i)$$

radios = 1.

Complex Conjugat.

$z = x + yi$

$z^* = z^\dagger = \bar{z} = x - iy \Rightarrow z^* = re^{-i\theta} \qquad zz^* = re^{i\theta}\cdot e^{-i\theta} = r^2, \quad r = \sqrt{z\cdot z^*}$

$z = r e^{i\theta}$ then $z^n = r^n \cdot e^{i n\theta} = r^n (\cos(n\theta) + i \sin(n\theta))$

$e^z = e^{x+iy} = e^x \cdot e^{iy}$ $\quad r = e^x$ and $\theta = y$

$e^{i\theta} = \cos\theta + i\sin\theta$

$e^{-i\theta} = \cos\theta - i\sin\theta$

$\cos\theta = \dfrac{e^{i\theta} + e^{-i\theta}}{2}$

$\sin\theta = \dfrac{e^{i\theta} - e^{-i\theta}}{2i} = i\left(\dfrac{e^{-i\theta} - e^{i\theta}}{2}\right)$

$(-i)$

a) $z^2 = e^{i 2n\pi}$

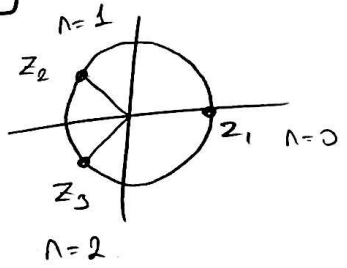
$z = e^{i\pi n}$ $\quad n = 0, 1, 2, 3 \cdots$ $\quad z_1 = 1$, $z_2 = -1$

2. Ders

b) $z^3 = 1$.

$z^3 = e^{i 2\pi n}$

$z = e^{i \frac{2}{3}\pi n}$



exercise

$z^n = -1$. find roots

$z^5 = -1$ "

Integrals

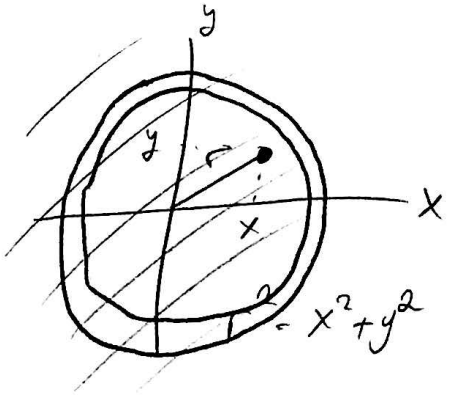
$$I = \int_{-\infty}^{+\infty} e^{-x^2} dx.$$

$$I^2 = \int_{-\infty}^{+\infty} e^{-x^2} dx \int_{-\infty}^{+\infty} e^{-y^2} dy = \iint_{-\infty-\infty}^{\infty\infty} e^{-(x^2+y^2)} \underbrace{dA}_{dx\,dy}$$

dV.

$$= \iint_{\mathbb{R}^2} e^{-r^2} dA_{ring} = \int_0^{+\infty} e^{-r^2} \cdot 2\pi r\, dr$$ / integration by part

exercise



The Area Ring $= 2\pi r dr$

$= \Pi$

$I = \sqrt{\pi}$

$$I(x) = \int_0^\infty e^{-y} \cdot y^{x-1}\, dy.$$

$$du = e^{-y} \cdot dy \quad , \quad u = -e^{-y}$$

$$dv = (x-1) y^{x-2} \cdot dy \quad , \quad v = y^{x-1}$$

$$I(x) = \underbrace{-y^{x-1} e^{-y} \Big|_0^\infty}_{0.} + \int_0^\infty (x-1)\, e^{-y} \cdot y^{x-2}\, dy =$$

$$= (x-1) \cdot \underbrace{\int_0^\infty e^{-y} \cdot y^{(x-2)} \cdot dy}_{I(x-1)} \Rightarrow I(x) = (x-1) \cdot I(x-1).$$

It is called Gamma function.

$$\boxed{\Gamma(x) = \int_0^\infty e^{-y} \cdot y^{x-1}\, dy}$$

For $x = 1$, $\Gamma(1) = 1$

$x = 2$, $\Gamma(2) = 1$

$\Gamma(3) = 2$

$\Gamma(4) = 3$

$$\boxed{\Gamma(n) = (n-1)!}$$

for integers $*$

Show that (Exercise).

$\Gamma(1/2) = \sqrt{\pi}$ using integration by parts.

Sphere in Geometry

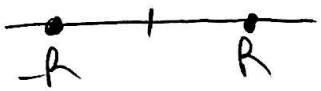
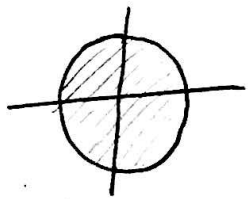
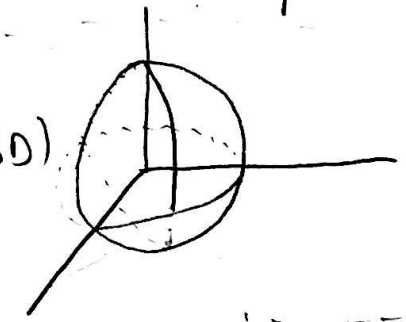
$R^2 = x^2 + y^2 + z^2$ yield collections of points on a 3D sphere

$R^2 \geq x^2 + y^2 + z^2$ " " " within 3D spherical surface.

Generalization of sphere in n-dim space with $x_1\ x_2 \ldots\ x_n$ coordinates

$x_1^2 + x_2^2 + x_3^2 + \ldots + x_n^2 = R^2$ corresponds to surface

$x_1^2 + x_2^2 + x_3^2 + \ldots + x_n^2 \leq R^2$ corresponds to volume

	equation	surface	volume	d
1D) 	$R^2 = x^2$	$S^{(1)}(R) = 2R^0$	$V^{(1)}(R) = 2R$	1
2D) 	$R^2 = x_1^2 + x_2^2$	$S^{(2)}(R) = 2\pi R$	$V^{(2)}(R) = \pi R^2$	2
3D) 	$R^2 = x_1^2 + x_2^2 + x_3^2$	$S^{(3)}(R) = 4\pi R^2$	$V^{(3)}(R) = \frac{4}{3}\pi R^3$	3